

Quantum State Preparation via Latent-Space Reinforcement Learning

A Physics-Informed World Model for Noisy Quantum Devices

Quantum Gym Project

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1 Problem Formulation

1.1 Quantum State and Target

We work with an $n = 3$ qubit system. The Hilbert space is

$$\mathcal{H} = (\mathbb{C}^2)^{\otimes 3}, \quad \dim \mathcal{H} = 8. \quad (1)$$

The target state is the Greenberger–Horne–Zeilinger (GHZ) state:

$$|\text{GHZ}\rangle = \frac{|000\rangle + |111\rangle}{\sqrt{2}}. \quad (2)$$

Starting from the computational zero state $|\psi_0\rangle = |000\rangle$, the goal is to find a parameterised circuit $U(\boldsymbol{\theta})$ such that

$$F(U(\boldsymbol{\theta})|000\rangle, |\text{GHZ}\rangle) = |\langle \text{GHZ} | U(\boldsymbol{\theta}) | 000 \rangle|^2 \geq 1 - \varepsilon. \quad (3)$$

1.2 Fubini–Study Metric

The natural distance on the space of pure quantum states (the projective Hilbert space $\mathbb{CP}^{2^n-1}$) is the *Fubini–Study metric*:

$$d_{\text{FS}}(|\psi\rangle, |\phi\rangle) = \arccos|\langle\psi|\phi\rangle| \in \left[0, \frac{\pi}{2}\right]. \quad (4)$$

We use the closely related *trace distance* as a scalar score:

$$D(\psi, \phi) = \frac{1}{2} \|\rho_\psi - \rho_\phi\|_1 = \sqrt{1 - |\langle\psi|\phi\rangle|^2} = \sin(d_{\text{FS}}(|\psi\rangle, |\phi\rangle)), \quad (5)$$

where $\rho_\psi = |\psi\rangle\langle\psi|$.

1.3 Key Requirement: Smooth Manifold

For the reinforcement learning agent to generalise, small angle increments must correspond to small displacements in $\mathcal{H}$. Given a layer parameterised by $\boldsymbol{\alpha}$ and perturbed by $\delta\boldsymbol{\alpha}$:

$$d_{\text{FS}}(U(\boldsymbol{\alpha})|\psi\rangle, U(\boldsymbol{\alpha} + \delta\boldsymbol{\alpha})|\psi\rangle) \propto \|\delta\boldsymbol{\alpha}\| \quad \text{for } \|\delta\boldsymbol{\alpha}\| \ll 1. \quad (6)$$

This is the motivating constraint for the ansatz design in Section 2.

2 Ansatz Design

2.1 Layer Structure

Each layer r of the forward ansatz applies:

$$L_r(\boldsymbol{\theta}_r) = \text{CR}(\alpha_{12}^{(r)}, 1 \rightarrow 2) \text{CR}(\alpha_{01}^{(r)}, 0 \rightarrow 1) \bigotimes_{q=0}^2 R_y(\theta_q^{(r)}), \quad (7)$$

where the parameter vector per layer is

$$\boldsymbol{\theta}_r = [\theta_0^{(r)}, \theta_1^{(r)}, \theta_2^{(r)}, \alpha_{01}^{(r)}, \alpha_{12}^{(r)}] \in \mathbb{R}^5. \quad (8)$$

Single-qubit gate

$$R_y(\theta) = e^{-i\frac{\theta}{2}Y} = \begin{pmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{pmatrix}. \quad (9)$$

Controlled- R_y (CR) gate

$$\text{CR}(\alpha) = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes R_y(\alpha) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \cos \frac{\alpha}{2} & -\sin \frac{\alpha}{2} \\ 0 & 0 & \sin \frac{\alpha}{2} & \cos \frac{\alpha}{2} \end{pmatrix}. \quad (10)$$

For $\alpha \rightarrow 0$, $\text{CR}(\alpha) \rightarrow I \otimes I$ continuously, ensuring smooth Hilbert-space displacements. The inverse is simply $\text{CR}(-\alpha)$.

2.2 Full Circuit

The N -layer forward circuit from $|000\rangle$ is:

$$|\psi_N\rangle = U_N(\Theta) |000\rangle, \quad U_N(\Theta) = \prod_{r=1}^N L_r(\theta_r), \quad (11)$$

where $\Theta = (\theta_1, \dots, \theta_N) \in \mathbb{R}^{5N}$.

2.3 Inverse Layer (Backward Pass)

Since $R_y^{-1}(\theta) = R_y(-\theta)$ and $\text{CR}^{-1}(\alpha) = \text{CR}(-\alpha)$, the exact inverse of L_r is:

$$L_r^{-1}(\theta_r) = \bigotimes_{q=0}^2 R_y(-\theta_q^{(r)}) \text{CR}(-\alpha_{01}^{(r)}, 0 \rightarrow 1) \text{CR}(-\alpha_{12}^{(r)}, 1 \rightarrow 2). \quad (12)$$

2.4 Hardware Quantization

The device (Quantum Inspire, Zebra/Tuna-17) quantizes angles to multiples of $\pi/36$ (5). Let $\Delta_{\text{device}} = \pi/36$. For the beam search we use a coarser step $\Delta_{\text{search}} = \pi/18$ (two device steps), chosen so that

$$\frac{\pi}{2} = 9 \times \frac{\pi}{18} = 9 \Delta_{\text{search}}, \quad (13)$$

meaning the optimal GHZ rotation angle is exactly reachable on the discrete grid.

3 Step 1: Beam Search in Simulation

3.1 Forward Search Objective

We want circuits that achieve high GHZ fidelity while keeping trajectories smooth. The beam search maximises:

$$\mathcal{F}(\Theta) = F(|\psi_N\rangle, |\text{GHZ}\rangle) = |\langle \text{GHZ} | U_N(\Theta) | 000 \rangle|^2. \quad (14)$$

3.2 Discrete Action Space

At each round r , the search perturbs the previous angles by:

$$\theta_r = \theta_{r-1} + \Delta\theta_r, \quad \Delta\theta_r \in \{-\Delta_{\text{search}}, 0, +\Delta_{\text{search}}\}^5. \quad (15)$$

This gives $3^5 = 243$ possible actions per step.

Algorithm 1 Forward Beam Search for GHZ Preparation

```

1: Initialise  $W$  beams with random  $\boldsymbol{\theta}_1 \sim \mathcal{U}(-\Delta, \Delta)^5$ 
2: for  $r = 1, \dots, N_{\max}$  do
3:   for each beam  $b$  with state  $|\psi_r^{(b)}\rangle$  do
4:     for each action  $\Delta\boldsymbol{\theta} \in \{-\Delta, 0, \Delta\}^5$  do
5:        $\boldsymbol{\theta}' \leftarrow \boldsymbol{\theta}_r^{(b)} + \Delta\boldsymbol{\theta}$ 
6:        $|\psi'\rangle \leftarrow L(\boldsymbol{\theta}')|\psi_r^{(b)}\rangle$ 
7:       Score:  $\mathcal{F}' \leftarrow |\langle \text{GHZ} | \psi' \rangle|^2$ 
8:     end for
9:   end for
10:  Keep top- $W$  candidates (80% greedy + 20% random exploration)
11:  if  $\max_b \mathcal{F}_r^{(b)} \geq 0.999$  then break
12:  end if
13: end for
14: return trajectories sorted by  $\mathcal{F}$  descending

```

3.3 Beam Search Algorithm

4 Step 2: Local Refinement

The discrete beam search is limited by the grid resolution Δ_{search} . Starting from the best beam trajectory $\boldsymbol{\Theta}^*$, we refine with continuous optimisation:

$$\boldsymbol{\Theta}^{**} = \arg \min_{\boldsymbol{\Theta}} [-\mathcal{F}(\boldsymbol{\Theta})] = \arg \min_{\boldsymbol{\Theta}} [1 - |\langle \text{GHZ} | U_N(\boldsymbol{\Theta}) | 000 \rangle|^2], \quad (16)$$

using L-BFGS-B with gradient computed via statevector simulation. This pushes $\mathcal{F} : 87\% \rightarrow 99\%+$.

After optimisation, the angles are re-quantized for hardware:

$$\tilde{\theta} = \Delta_{\text{device}} \cdot \text{round}\left(\frac{\theta^{**}}{\Delta_{\text{device}}}\right). \quad (17)$$

5 Step 3: Shadow Fingerprinting and QI Acquisition

5.1 Classical Shadow Fingerprint

To encode a quantum state $|\psi\rangle$ into a fixed-dimensional feature vector, we use *classical shadow tomography*. For $n = 3$ qubits, a shadow of size K gives an unbiased estimator for all k -local Pauli observables ($k \leq 3$).

The fingerprint $\mathbf{x}(\psi) \in \mathbb{R}^{63}$ contains expectation values of:

- 1-local Paulis: $\{X_i, Y_i, Z_i\}$ for $i \in \{0, 1, 2\}$ — 9 values
- 2-local Paulis: $\{P_i P_j\}$ for $i < j$, $P \in \{X, Y, Z\}$ — 27 values
- 3-local Paulis: $\{P_0 P_1 P_2\}$ — 27 values

Total: $9 + 27 + 27 = 63$ features.

Each estimator from K shots satisfies:

$$\hat{o}_P = \mathbb{E}_{\text{shadow}}[\text{tr}(P \hat{\rho})] = \text{tr}(P \rho_\psi), \quad \text{Var}[\hat{o}_P] = \mathcal{O}(K^{-1}). \quad (18)$$

5.2 Budget-Aware QI Circuit Selection

Running circuits on QI is expensive (queue time). We select $M \approx 50$ circuits maximising information gain subject to a budget:

$$\max_{\mathcal{S} \subseteq \mathcal{C}, |\mathcal{S}| \leq M} \text{InfoGain}(\mathcal{S}), \quad (19)$$

where the selected set $\mathcal{S}$ is constructed greedily:

Priority 1 — Anchor points (~ 10 circuits):

$$\mathcal{S}_1 = \{|000\rangle, |\text{GHZ}\rangle, |\psi_{r^*}\rangle \text{ for } r^* \in \{1, 3, 5, 7, 9\}\}. \quad (20)$$

Priority 2 — Noise characterization (~ 20 circuits): Single-gate benchmarks $\{R_y(\theta)|0\rangle\}_\theta$ and $\{\text{CR}(\alpha)|10\rangle\}_\alpha$ to fit a noise model.

Priority 3 — Trajectory diversity (~ 20 circuits): Sparse steps from the top-3 beam trajectories.

6 Step 4: Variational Autoencoder (VAE)

6.1 Architecture

The VAE encodes shadow fingerprints $\mathbf{x} \in \mathbb{R}^{63}$ to a low-dimensional latent space $\mathbf{z} \in \mathbb{R}^d$ ($d = 2$ for visualization).

The encoder outputs a Gaussian posterior:

$$q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}\left(\boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag}\left(e^{\boldsymbol{\sigma}_\phi^2(\mathbf{x})}\right)\right). \quad (21)$$

The decoder reconstructs the fingerprint:

$$p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\hat{\mathbf{x}}_\theta(\mathbf{z}), \sigma_{\text{dec}}^2 I). \quad (22)$$

6.2 Standard VAE Loss

$$\mathcal{L}_{\text{VAE}}(\phi, \theta; \mathbf{x}) = \underbrace{\mathbb{E}_{q_\phi}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{reconstruction}} - \underbrace{\beta D_{\text{KL}}[q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})]}_{\text{regularisation}}, \quad (23)$$

where $p(\mathbf{z}) = \mathcal{N}(0, I)$ is the prior and β controls the disentanglement–fidelity trade-off via KL annealing:

$$\beta(e) = \beta_{\min} + (\beta_{\max} - \beta_{\min}) \cdot \min\left(1, \frac{e}{E_{\text{warmup}}}\right). \quad (24)$$

6.3 Metric Loss (Physics-Informed)

To ensure the latent geometry reflects the Fubini–Study metric, we add:

$$\mathcal{L}_{\text{metric}} = \sum_{(i,j) \in \mathcal{T}} \left(\|\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\|^2 - \lambda d_{\text{FS}}^2(|\psi_i\rangle, |\psi_j\rangle) \right)^2, \quad (25)$$

where $\mathcal{T}$ is the set of consecutive transition pairs. This forces nearby states in $\mathcal{H}$ to be nearby in the latent space.

6.4 Sim-to-Real Fine-Tuning

After training on simulation data, the encoder ϕ is fine-tuned on QI fingerprints $\{\mathbf{x}_j^{\text{QI}}\}$ with the decoder frozen (θ fixed):

$$\phi^* = \arg \min_{\phi} \sum_j \mathcal{L}_{\text{VAE}}(\phi, \theta_{\text{frozen}}; \mathbf{x}_j^{\text{QI}}), \quad (26)$$

using a small learning rate to preserve the latent geometry while aligning QI points correctly.

6.5 Joint Training Objective

The full training loss is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{VAE}} + \lambda_1 \mathcal{L}_{\text{transition}} + \lambda_2 \mathcal{L}_{\text{metric}} + \lambda_3 \mathcal{L}_{\text{noise}}, \quad (27)$$

with $\mathcal{L}_{\text{transition}}$ defined in Section 7.

7 Step 5: Stochastic World Model

7.1 Transition Dynamics

The world model predicts the next latent state given the current state and action:

$$\boldsymbol{\mu}_{t+1} = f_{\omega}(\boldsymbol{\mu}_t, \mathbf{a}_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \text{diag}(\boldsymbol{\sigma}_{\omega}^2(\boldsymbol{\mu}_t, \mathbf{a}_t))), \quad (28)$$

where the action is the angle delta vector:

$$\mathbf{a}_t = \Delta \boldsymbol{\theta}_t \in \{-\Delta_{\text{search}}, 0, +\Delta_{\text{search}}\}^5 \subset \mathbb{R}^5. \quad (29)$$

7.2 Residual Architecture

Rather than predicting $\boldsymbol{\mu}_{t+1}$ directly, the model predicts the *displacement* $\Delta \boldsymbol{\mu}$:

$$\Delta \boldsymbol{\mu} = g_{\omega}(\boldsymbol{\mu}_t, \mathbf{a}_t), \quad \hat{\boldsymbol{\mu}}_{t+1} = \boldsymbol{\mu}_t + \Delta \boldsymbol{\mu}. \quad (30)$$

This residual structure enforces:

- **Zero-action consistency:** $g_{\omega}(\boldsymbol{\mu}, \mathbf{0}) \approx \mathbf{0}$
- **Smoothness:** $\|\Delta \boldsymbol{\mu}\| \propto \|\mathbf{a}_t\|$ for small actions
- **Generalization:** Local geometry is learned, not global positions

7.3 Heteroscedastic Noise Model

The state-dependent noise $\boldsymbol{\sigma}_{\omega}^2(\boldsymbol{\mu}_t, \mathbf{a}_t)$ encodes hardware imperfections. We train it to match the empirical variance between simulation and QI measurements:

$$\boldsymbol{\sigma}_{\text{target}}^2(\boldsymbol{\mu}_t, \mathbf{a}_t) = \left\| \boldsymbol{\mu}_{t+1}^{\text{ideal}} - \boldsymbol{\mu}_{t+1}^{\text{QI}} \right\|^2. \quad (31)$$

The transition loss combines MSE on the mean with negative log-likelihood for the variance:

$$\mathcal{L}_{\text{transition}} = \underbrace{\left\| \hat{\boldsymbol{\mu}}_{t+1} - \boldsymbol{\mu}_{t+1} \right\|^2}_{\text{mean MSE}} + \underbrace{\frac{1}{2} \sum_i \left[\frac{(\hat{\mu}_{t+1,i} - \mu_{t+1,i})^2}{\sigma_i^2} + \log \sigma_i^2 \right]}_{\text{NLL (noise)}}. \quad (32)$$

The noise loss trained on QI data:

$$\mathcal{L}_{\text{noise}} = \sum_{j \in \mathcal{S}_{\text{QI}}} \left\| \boldsymbol{\sigma}_{\omega}^2(\boldsymbol{\mu}_j, \mathbf{a}_j) - \left(\boldsymbol{\mu}_{j+1}^{\text{ideal}} - \boldsymbol{\mu}_{j+1}^{\text{QI}} \right)^2 \right\|^2. \quad (33)$$

7.4 Data Augmentation

For each transition $(\theta_t, \mathbf{a}_t)$ in the beam search dataset, we generate K_{aug} synthetic neighbours by perturbing angles:

$$\tilde{\theta}_t = \theta_t + \xi, \quad \xi \sim \mathcal{U}(-\Delta_{\text{search}}, +\Delta_{\text{search}})^5, \quad (34)$$

giving $|\mathcal{D}| \times K_{\text{aug}}$ additional transitions at negligible cost.

8 Step 6: Reinforcement Learning in Latent Space

8.1 MDP Formulation

The RL problem is defined as a Markov Decision Process:

$$\text{State: } s_t = \boldsymbol{\mu}_t \in \mathbb{R}^d \quad (35)$$

$$\text{Action: } a_t \in \mathcal{A} = \{-\Delta, 0, +\Delta\}^5, \quad |\mathcal{A}| = 3^5 = 243 \quad (36)$$

$$\text{Transition: } s_{t+1} \sim p_{\omega}(s_{t+1}|s_t, a_t) = \mathcal{N}(\hat{\boldsymbol{\mu}}_{t+1}, \boldsymbol{\sigma}_{\omega}^2) \quad (37)$$

$$\text{Reward: } r_t = -D(|\psi_t\rangle, |\text{GHZ}\rangle) + \mathcal{K}[D_t < \varepsilon] \cdot R_{\text{terminal}} \quad (38)$$

$$\text{Horizon: } T = N_{\text{max}} \text{ rounds} \quad (39)$$

The reward function explicitly penalises distance to GHZ:

$$r_t = \underbrace{F_t - F_{t-1}}_{\text{fidelity gain}} - \underbrace{\lambda_{\text{step}}}_{\text{step penalty}} + \underbrace{R_T \cdot \mathcal{K}[F_T \geq F^*]}_{\text{terminal bonus}}, \quad (40)$$

where $F_t = |\langle \text{GHZ} | \psi_t \rangle|^2$ and $F^* = 0.99$.

8.2 Uncertainty-Penalised Reward (Pessimistic RL)

On the real device, regions with high model uncertainty should be avoided. We augment the reward with an epistemic penalty:

$$r_t^{\text{pessimistic}} = r_t - \kappa \|\boldsymbol{\sigma}_{\omega}(s_t, a_t)\|, \quad (41)$$

where $\kappa > 0$ controls risk aversion. This keeps the RL policy within the *trust region* of the world model — states well-covered by training data.

8.3 Policy Gradient Objective

We train a policy $\pi_{\psi}(a_t|s_t)$ using proximal policy optimisation (PPO):

$$\mathcal{L}_{\text{PPO}}(\psi) = -\mathbb{E}_t \left[\min \left(\frac{\pi_{\psi}(a_t|s_t)}{\pi_{\psi_{\text{old}}}(a_t|s_t)} \hat{A}_t, \text{clip} \left(\frac{\pi_{\psi}}{\pi_{\psi_{\text{old}}}}, 1 - \epsilon_{\text{clip}}, 1 + \epsilon_{\text{clip}} \right) \hat{A}_t \right) \right], \quad (42)$$

where the advantage estimate is:

$$\hat{A}_t = \sum_{k=0}^{T-t} (\gamma\lambda)^k (r_{t+k} + \gamma V(s_{t+k+1}) - V(s_{t+k})). \quad (43)$$

9 Step 7: Execution on Hardware

9.1 Policy Rollout

The trained policy π^* is executed greedily:

$$a_t^* = \arg \max_a \pi^*(a|\boldsymbol{\mu}_t), \quad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + a_t^*. \quad (44)$$

The circuit $U_T(\boldsymbol{\Theta}^*)$ is compiled to hardware-native gates with angle quantization $\Delta_{\text{device}} = \pi/36$.

9.2 Online Fingerprinting

After each executed layer r , the device returns measurement outcomes from K shots. The shadow fingerprint $\hat{\mathbf{x}}_r^{\text{QI}}$ is computed and fed to the VAE encoder:

$$\hat{\boldsymbol{\mu}}_r = \boldsymbol{\mu}_\phi(\hat{\mathbf{x}}_r^{\text{QI}}), \quad (45)$$

giving real-time feedback on the actual quantum state.

9.3 Closed-Loop Correction

If the measured latent state deviates from the predicted trajectory:

$$\|\hat{\boldsymbol{\mu}}_r - \hat{\boldsymbol{\mu}}_r^{\text{predicted}}\| > \delta_{\text{tol}}, \quad (46)$$

the policy is re-queried with $\hat{\boldsymbol{\mu}}_r$ as the new starting state, allowing adaptive correction of hardware errors.

10 Generalisation of the World Model

10.1 The Core Challenge

The world model is trained on ~ 1000 transitions covering a small region of $\mathcal{H}$, but must generalise to unseen transitions. Three complementary strategies:

10.1.1 1. Physics Inductive Bias (Residual + Smoothness)

The residual parameterisation $\hat{\boldsymbol{\mu}}_{t+1} = \boldsymbol{\mu}_t + g_\omega(\boldsymbol{\mu}_t, \mathbf{a}_t)$ combined with the smoothness constraint

$$\mathcal{L}_{\text{smooth}} = \sum_t \left(\frac{\|g_\omega(\boldsymbol{\mu}_t, \mathbf{a}_t)\|}{\|\mathbf{a}_t\|} - \lambda_{\text{smooth}} \right)^2 \quad (47)$$

ensures that the model extrapolates smoothly near training data.

10.1.2 2. Symmetry Exploitation (GNN)

The qubit topology $(0) - (1) - (2)$ has a message-passing symmetry. A Graph Neural Network with node features $h_q = (\theta_q, \mu_q^{\text{local}})$ and edge features $e_{qr} = \alpha_{qr}$ performs:

$$h'_q = \sigma \left(W_{\text{self}} h_q + \sum_{r \in \mathcal{N}(q)} W_{\text{msg}} [h_r \| e_{qr}] \right), \quad (48)$$

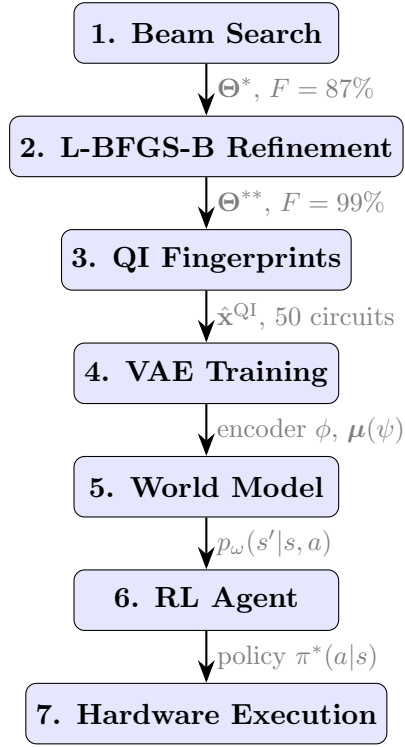
sharing parameters across edges $(0, 1)$ and $(1, 2)$, effectively doubling training data.

10.1.3 3. Metric Contrastive Loss (Triplet)

$$\mathcal{L}_{\text{triplet}} = \sum_t \max \left(0, \|\boldsymbol{\mu}_t - \boldsymbol{\mu}_{t+1}\|^2 - \|\boldsymbol{\mu}_t - \boldsymbol{\mu}_{\text{neg}}\|^2 + m \right), \quad (49)$$

where $\boldsymbol{\mu}_{\text{neg}}$ is a randomly sampled latent state and $m > 0$ is the margin. This enforces that the latent geometry is consistent with quantum neighborhood structure, enabling generalization to unseen regions.

11 Full Pipeline Summary



Step	Input	Output	Cost
1. Beam Search	Ansatz, Δ_s	Top trajectories Θ^*	Cheap (sim)
2. Refinement	Θ^*	$\Theta^{**}, F \geq 99\%$	Cheap (sim)
3. QI Acquisition	Θ^{**}	$\{\hat{\mathbf{x}}_j^{\text{QI}}\}$	50 QI runs
4. VAE	$\hat{\mathbf{x}}$	Latent μ , encoder ϕ	GPU hours
5. World Model	$(\mu_t, \mathbf{a}_t, \mu_{t+1})$	f_ω, σ_ω	GPU hours
6. RL	f_ω	Policy π^*	GPU hours
7. Hardware	π^*	GHZ state, F_{hw}	QI runs

Table 1: Pipeline summary. QI = Quantum Inspire hardware runs.